

Linear Control Design 1 Exam Formula Sheet

34721/34722 Linear Control Design 1

Spring 2026

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1 Exam map and control concepts

1.1 Ctrl+F keywords

1.1.1 Course keywords, eksamen, regulering, feedback, controller

$$\{r, e, u, y, y_m, G, C, H, L, S, T, \omega_c, \gamma_M, A_M, e_{ss}\}$$

Use this file as the front-door map. Danish search aliases used below include regulering, tilbagekobling, forstyrrelse, maaling, folgeføj, stabilitet, and bodeplot.

1.2 Signal and loop notation

1.2.1 Standard feedback signals

$$e(t) = r(t) - y_m(t), \quad y_m(t) = H(s)y(t), \quad u(t) = C(s)e(t)$$

r is reference, e is error, u is plant input, y is output, and y_m is measured output.

1.2.2 Plant, controller, measurement, loop transfer

$$L(s) = C(s)G(s)H(s)$$

Most stability-margin and sensitivity questions start by identifying the correct loop transfer function L .

1.2.3 Closed-loop transfer and characteristic equation

$$T(s) = \frac{L(s)}{1 + L(s)}, \quad 1 + L(s) = 0$$

Closed-loop stability is decided by the roots of the characteristic equation, not by open-loop poles alone.

1.3 Exam problem recognition

1.3.1 Problem type keyword table

Exam wording	Go to
block diagram, feedback, summing junction	Block diagrams and feedback
RLC, masses, equations of motion, model	Modelling and linearization
differential equation, Laplace, poles, zeros	Laplace transform and transfer functions
step response, overshoot, steady-state value	Frequency and time-domain analysis
Bode, gain margin, phase margin, K_P	Bode plots and stability margins
Nyquist, encirclement, unstable open loop	Nyquist plot and unstable systems
PI-Lead, Lead, Lag, α, β, N_i	PI-Lead design and limited systems
disturbance, sensitivity, prefilter, feed-forward	Disturbances and feed-forward

Use the wording in the problem statement first, then confirm with the formulas in the relevant section.

1.4 Controller templates

1.4.1 P, PI, Lead, Lag controller forms

$$\begin{aligned} C_P(s) &= K_P, \\ C_{PI}(s) &= K_P \frac{\tau_i s + 1}{\tau_i s}, \\ C_{\text{lead}}(s) &= \frac{\tau_d s + 1}{\alpha \tau_d s + 1}, \quad 0 < \alpha < 1, \\ C_{\text{lag}}(s) &= \frac{\tau_i s + 1}{\frac{\tau_i}{\beta} s + 1}, \quad \beta > 1. \end{aligned}$$

These are the recurring controller blocks in the lectures and F21 exam.

1.4.2 Performance and robustness symbols

t_r	rise time
t_s	settling time
M_p	percent overshoot
e_{ss}	steady-state error, stationær fejl
ω_c	gain crossover frequency
γ_M	phase margin
A_M	gain margin

Fast exam checks often use signs, units, limits, and whether these specifications move in the expected direction.

1.5 Multiple-choice and fast checks

1.5.1 Elimination rules for answer options

wrong sign $\vee$ wrong units $\vee$ unstable hidden pole $\vee$ improper controller $\implies$ reject option

For multi-clause answers, one false clause makes the option false.

1.5.2 Limit checks

$$\lim_{K_P \rightarrow 0} T(s) = 0, \quad \lim_{K_P \rightarrow \infty} \frac{K_P G(s)}{1 + K_P G(s)} \approx 1$$

Use limiting cases to reject transfer functions with impossible signs or missing feedback denominators.

2 Block diagrams and feedback

2.1 Ctrl+F keywords

2.1.1 Block diagram, blokdiagram, feedback, tilbagekobling, summing junction

series, parallel, feedback, nested loop, non-unity feedback, error, folgeføj

Use this section when a problem shows a block diagram or asks for Y/R , E/R , or steady-state error from a diagram.

2.2 Core block reductions

2.2.1 Series and parallel blocks

$$G_{\text{series}} = G_1 G_2, \quad G_{\text{parallel}} = G_1 + G_2$$

Series paths multiply. Parallel paths with the same input and summed output add with the signs shown at the summing junction.

2.2.2 Negative feedback with measurement block

$$\frac{Y(s)}{R(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}, \quad \frac{E(s)}{R(s)} = \frac{1}{1 + C(s)G(s)H(s)}$$

The feedback denominator is $1 + L(s)$ for negative feedback with $L = CGH$.

2.2.3 Positive feedback sign trap

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

Change only the sign required by the summing junction. Many wrong options use $1 + GH$ where the diagram requires $1 - GH$.

2.2.4 Unity feedback

$$T(s) = \frac{Y}{R} = \frac{L}{1 + L}, \quad S(s) = \frac{E}{R} = \frac{1}{1 + L}$$

For unity feedback $L = CG$. These two formulas also identify tracking and sensitivity paths.

2.3 Exam recipe: reduce a block diagram

Use when: the problem asks for equivalent transfer function, closed-loop transfer, or block diagram answer option.

Steps:

1. Label each summing junction equation before simplifying.
2. Collapse obvious series and parallel blocks.
3. Reduce innermost feedback loops first.
4. Keep signs from the summing junctions, not from memory.
5. Check the final denominator for the expected $1 \pm L$ structure.

Fast checks: set one block equal to zero or one and verify the simplified diagram still makes sense.

Common traps: moving a summing junction across a block without multiplying or dividing by that block; losing measurement block $H(s)$.

2.4 Exam recipe: non-unity feedback steady-state error

Use when: the problem includes a feedback branch gain/filter and asks for final error.

Steps:

1. Derive $E(s)/R(s)$ directly from the diagram.
2. Insert the reference input, usually $R(s) = 1/s$ for a unit step.
3. Use $e_{ss} = \lim_{s \rightarrow 0} sE(s)$ only after checking closed-loop stability.
4. Evaluate all stable transfer functions at $s = 0$.

Fast checks: if $L(0)$ is large, error should be small; if a non-unity sensor has DC gain $H(0) \neq 1$, the controlled output may not equal the reference.

Common traps: using $1/(1 + CG)$ when the diagram has $H(s)$, extra branches, or nested loops.

2.5 Controller blocks from diagrams

2.5.1 P, PI, PD, PID in time and Laplace form

$$\begin{aligned}C_P(s) &= K_P, \\C_{PI}(s) &= K_P \frac{\tau_i s + 1}{\tau_i s}, \\C_{PD}(s) &= K_P(1 + \tau_d s), \\C_{PID}(s) &= K_P \left(1 + \frac{1}{\tau_i s} + \tau_d s\right).\end{aligned}$$

Use the controller form exactly as shown in the block diagram or problem statement.

3 Laplace transform and transfer functions

3.1 Ctrl+F keywords

3.1.1 Laplace, differential equation, transfer function, poles, zeros, poler, nulpunkter

ODE, characteristic equation, zero initial conditions, proper, strictly proper

Use this section when an exam problem gives a differential equation or asks for poles, zeros, stability, or $G(s)$.

3.2 Laplace rules

3.2.1 Unilateral Laplace transform

$$X(s) = \mathcal{L}\{x(t)\} = \int_{0^-}^{\infty} x(t)e^{-st} dt$$

The transform converts linear differential equations into algebraic equations.

3.2.2 Derivative property with initial conditions

$$\mathcal{L}\{x^{(n)}(t)\} = s^n X(s) - \sum_{k=0}^{n-1} s^{n-1-k} x^{(k)}(0^-)$$

For transfer functions, set all initial conditions to zero.

3.2.3 Common transform pairs

$x(t)$	$X(s)$
$\delta(t)$	1
$u(t)$	$1/s$
$tu(t)$	$1/s^2$
$e^{-at}u(t)$	$1/(s+a)$
$\sin(\omega t)u(t)$	$\omega/(s^2 + \omega^2)$
$\cos(\omega t)u(t)$	$s/(s^2 + \omega^2)$

These pairs are enough for the recurring transfer-function and final-value tasks.

3.3 Core transfer-function formulas

3.3.1 Transfer function definition

$$G(s) = \left. \frac{Y(s)}{U(s)} \right|_{\text{zero initial conditions}}$$

A transfer function is only the input-output relation for the selected input and output.

3.3.2 Differential equation to transfer function

$$a_n y^{(n)} + \dots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + \dots + b_1 \dot{u} + b_0 u$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \dots + b_1 s + b_0}{a_n s^n + \dots + a_1 s + a_0}$$

Denominator roots are poles. Numerator roots are zeros.

3.3.3 Poles, zeros, gain form

$$G(s) = K \frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}$$

Poles decide natural modes and stability. Zeros shape response and Bode phase.

3.3.4 Properness and implementability

$$\deg N(s) \leq \deg D(s) \quad \text{proper}, \quad \deg N(s) < \deg D(s) \quad \text{strictly proper}$$

Controllers, filters, and feed-forward blocks must be proper for direct implementation.

3.4 Exam recipe: ODE to transfer function and poles

Use when: the problem gives a differential equation and asks for $G(s)$, poles, stability, or a multiple-choice transfer function.

Steps:

1. Set initial conditions to zero.
2. Replace $y^{(k)}$ by $s^k Y(s)$ and $u^{(k)}$ by $s^k U(s)$.
3. Move all Y -terms to one side and U -terms to the other.
4. Form $G(s) = Y(s)/U(s)$.
5. Factor or inspect the denominator for poles.

Fast checks: highest derivative of y sets denominator order; signs in the ODE carry into the polynomial.

Common traps: keeping nonzero initial-condition terms inside a transfer function; confusing poles of G with zeros of G .

3.5 Exam recipe: characteristic equation from feedback

Use when: the problem asks for closed-loop poles or stability of a feedback system.

Steps:

1. Find the loop transfer $L(s)$.
2. Write the characteristic equation $1 + L(s) = 0$ for negative feedback.
3. Clear denominators to get a polynomial.
4. Test pole locations or use margin/Nyquist methods when the polynomial is not requested.

Fast checks: numerator zeros of $T = L/(1 + L)$ do not decide closed-loop stability unless they cancel unstable poles, which should not be assumed.

4 Frequency and time-domain analysis

4.1 Ctrl+F keywords

4.1.1 Step response, overshoot, damping, steady-state value, stationaer vaerdi

DC gain, final value theorem, sinusoidal response, damping ratio, settling time

Use this section for time-response questions and for checking whether a Bode or controller answer gives plausible dynamics.

4.2 Steady-state and final value

4.2.1 DC gain

$$K_{ss} = G(0) = \lim_{s \rightarrow 0} G(s)$$

For a stable system with constant input, K_{ss} is the output/input ratio.

4.2.2 Final value theorem

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

All poles of $sY(s)$ must be in the open left half-plane for this theorem to be valid.

4.3 Frequency response

4.3.1 Evaluate transfer function on the imaginary axis

$$G(j\omega) = G(s)|_{s=j\omega}$$

Magnitude and phase describe sinusoidal steady-state behavior.

4.3.2 Sinusoidal steady-state output

$$u(t) = A \sin(\omega t) \implies y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$$

The magnitude scales the amplitude and the phase shifts the sinusoid.

4.4 First-order response

4.4.1 First-order transfer function and step response

$$G(s) = \frac{K}{\tau s + 1}, \quad y(t) = K \left(1 - e^{-t/\tau}\right) u(t)$$

At $t = \tau$, the response has reached 63.2% of the total change.

4.5 Second-order response

4.5.1 Standard second-order denominator

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Compare a closed-loop denominator with this form to identify ζ and ω_n .

4.5.2 Second-order poles

$$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Complex poles occur for $0 < \zeta < 1$ and produce oscillation and overshoot.

4.5.3 Percent overshoot

$$M_p = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}\%$$

Use this for underdamped second-order unit-step specifications.

4.5.4 Settling time approximation

$$t_s \approx \frac{4}{\zeta\omega_n}$$

This is the common 2% settling-time approximation.

4.6 Stability from poles

4.6.1 Asymptotic stability

$$\operatorname{Re}(p_k) < 0 \quad \text{for all poles } p_k$$

All poles must lie in the open left half-plane.

4.6.2 Unstable and marginal pole cases

$$\exists p_k : \operatorname{Re}(p_k) > 0 \Rightarrow \text{unstable}, \quad \text{repeated } p_k = 0 \Rightarrow \text{unstable}$$

Simple imaginary-axis poles are marginal only in special cases; repeated poles at the origin are not stable.

4.7 Exam recipe: overshoot limit from closed-loop denominator

Use when: the problem asks whether a K_P , ζ , or closed-loop transfer function satisfies an overshoot bound.

Steps:

1. Put the denominator in $s^2 + 2\zeta\omega_n s + \omega_n^2$ form.
2. Solve for ω_n from the constant term.
3. Solve for ζ from the s -coefficient.
4. Insert ζ into $M_p = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}\%$.
5. Compare with the required overshoot.

Fast checks: larger ζ means lower overshoot; $\zeta \geq 1$ gives no oscillatory overshoot in the standard model.

Common traps: applying the formula to a higher-order response without checking dominant second-order behavior.

4.8 Exam recipe: final value and steady-state output

Use when: the problem asks for final output, DC gain, or steady-state response.

Steps:

1. Write $Y(s) = G(s)U(s)$ or the closed-loop transfer from the diagram.
2. Insert the input transform, often $1/s$ for a unit step.
3. Check the poles of $sY(s)$ are stable.
4. Compute $\lim_{s \rightarrow 0} sY(s)$.

Fast checks: for a stable step response, the answer should match DC gain times step amplitude.

5 Modelling and linearization

5.1 Ctrl+F keywords

5.1.1 RLC, impedance, mass spring damper, equations of motion, modelling

Newton, force balance, voltage divider, white-box model, linearization, operating point

Use this section when the exam asks for a model before doing control design.

5.2 Mechanical modelling

5.2.1 Newton's second law

$$\sum F = m\ddot{x}$$

Choose a positive direction and keep spring, damper, and external-force signs consistent.

5.2.2 Spring and damper forces

$$F_k = k(x_1 - x_2), \quad F_b = b(\dot{x}_1 - \dot{x}_2)$$

For coupled masses, relative displacement and relative velocity determine internal forces.

5.2.3 Mass-spring-damper transfer function

$$m\ddot{x} + b\dot{x} + kx = F \implies \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

With zero initial conditions, force-balance equations become algebraic transfer functions.

5.3 Electrical modelling

5.3.1 Laplace impedances

$$Z_R = R, \quad Z_L = sL, \quad Z_C = \frac{1}{sC}$$

Use impedances for RLC circuit transfer functions in the s -domain.

5.3.2 Voltage divider

$$\frac{V_2(s)}{V_{\text{in}}(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

Pick Z_2 as the impedance across the measured output voltage.

5.4 Linearization

5.4.1 Equilibrium point

$$0 = f(x_0, u_0), \quad y_0 = g(x_0, u_0)$$

The operating point must satisfy the nonlinear state equation for constant input.

5.4.2 First-order Taylor linearization

$$\begin{aligned} \Delta \dot{x} &= A\Delta x + B\Delta u, & \Delta y &= C\Delta x + D\Delta u \\ A &= \left. \frac{\partial f}{\partial x} \right|_0, & B &= \left. \frac{\partial f}{\partial u} \right|_0, & C &= \left. \frac{\partial g}{\partial x} \right|_0, & D &= \left. \frac{\partial g}{\partial u} \right|_0 \end{aligned}$$

The variables are deviations from the operating point.

5.4.3 State-space to transfer function

$$G(s) = C(sI - A)^{-1}B + D$$

Use this only after selecting the correct input and output channels.

5.5 Exam recipe: RLC circuit transfer function

Use when: the problem shows resistors, inductors, capacitors, and asks for $V_{\text{out}}/V_{\text{in}}$ or a block diagram.

Steps:

1. Replace R, L, C by $R, sL, 1/(sC)$.
2. Identify series and parallel equivalent impedances.
3. Use the voltage divider or Kirchhoff equations.
4. Simplify to a rational function in s .
5. Check low- and high-frequency limits.

Fast checks: capacitors are open at DC and short at high frequency; inductors are short at DC and open at high frequency.

Common traps: choosing the wrong output impedance in the voltage divider.

5.6 Exam recipe: coupled mass equations of motion

Use when: the problem asks for equations of motion or transfer function of masses, springs, and dampers.

Steps:

1. Draw one free-body equation for each mass.
2. Use relative displacement for springs and relative velocity for dampers.
3. Apply $\sum F = m\ddot{x}$ to each mass.
4. Laplace transform with zero initial conditions.
5. Solve the algebraic equations for the requested output/input ratio.

Fast checks: internal spring and damper forces appear with opposite signs on the two connected masses.

6 Bode plots and stability margins

6.1 Ctrl+F keywords

6.1.1 Bode plot, bodeplot, gain margin, phase margin, crossover, stability margin

magnitude, dB, slope, poles, zeros, $K_P, \omega_c, \omega_\pi, \gamma_M, A_M$

Use this section for the most common F21 tasks: read a Bode plot, infer $G(s)$, choose K_P , and check stability margins.

6.2 Bode basics

6.2.1 Magnitude in decibels

$$M_{\text{dB}} = 20 \log_{10} |G(j\omega)|, \quad |G(j\omega)| = 10^{M_{\text{dB}}/20}$$

Products become sums in dB. A proportional gain shifts only the magnitude curve.

6.2.2 Bode factor form

$$G(s) = K \frac{\prod_i (1 + s/\omega_{z_i})}{s^q \prod_k (1 + s/\omega_{p_k})}$$

Break frequencies are visible as slope changes in the magnitude plot.

6.2.3 Real pole, real zero, integrator rules

factor	Δslope	Δphase
$1/(1 + s/\omega_p)$	-20 dB/dec	-90°
$1 + s/\omega_z$	$+20 \text{ dB/dec}$	$+90^\circ$
$1/s$	-20 dB/dec	-90°

Use slope changes to count poles and zeros.

6.2.4 Right-half-plane zero and pole phase trap

$$1 - \frac{s}{\omega_z} : \quad \Delta M = +20 \text{ dB/dec}, \quad \Delta\phi = -90^\circ,$$

$$\frac{1}{1 - s/\omega_p} : \quad \Delta M = -20 \text{ dB/dec}, \quad \Delta\phi = +90^\circ.$$

Right-half-plane factors have the usual magnitude slope. Their phase contribution is reversed compared with stable factors.

6.3 Stability margins

6.3.1 Gain crossover and phase margin

$$|L(j\omega_c)| = 1, \quad \gamma_M = 180^\circ + \angle L(j\omega_c)$$

ω_c is where the open-loop magnitude crosses 0 dB.

6.3.2 Phase crossover and gain margin

$$\angle L(j\omega_\pi) = -180^\circ, \quad A_M = \frac{1}{|L(j\omega_\pi)|}, \quad A_{M,\text{dB}} = -20 \log_{10} |L(j\omega_\pi)|$$

Gain margin is read at the -180° phase crossing.

6.3.3 P-controller gain shift

$$L(s) = K_P G(s), \quad 20 \log_{10} |L| = 20 \log_{10} K_P + 20 \log_{10} |G|$$

Positive K_P shifts magnitude but does not change phase.

6.4 Exam recipe: infer transfer function from Bode plot

Use when: the problem gives magnitude/phase plots and answer options for $G(s)$.

Steps:

1. Read initial low-frequency slope to count integrators.
2. Mark each break frequency where slope changes.
3. Slope change $+20$ means one zero; -20 means one pole.
4. Use phase to distinguish stable, unstable, and right-half-plane factors.

5. Use one magnitude point, often DC gain or $\omega = 1$, to find K .

Fast checks: denominator order minus numerator order equals final negative slope divided by 20 dB/dec.

Common traps: missing double poles/zeros; ignoring phase when a right-half-plane zero has the same magnitude as a left-half-plane zero.

6.5 Exam recipe: choose K_P for desired phase margin

Use when: the problem asks for proportional gain that gives a specified γ_M .

Steps:

1. Find the frequency where $\angle G(j\omega) = -180^\circ + \gamma_M$.
2. Read $M_{dB} = 20 \log_{10} |G(j\omega)|$ at that frequency.
3. Choose $20 \log_{10} K_P = -M_{dB}$.
4. Convert $K_P = 10^{-M_{dB}/20}$.

Fast checks: if the plant magnitude is below 0 dB, $K_P > 1$; if it is above 0 dB, $K_P < 1$.

Common traps: using the current 0 dB crossing instead of the frequency set by the desired phase.

6.6 Exam recipe: P-controller stability range from Bode plot

Use when: the problem asks for the range of K_P giving closed-loop stability.

Steps:

1. Find ω_π where $\angle G(j\omega_\pi) = -180^\circ$.
2. Read $|G(j\omega_\pi)|$ or dB magnitude.
3. Critical gain is $K_{crit} = 1/|G(j\omega_\pi)|$.
4. For stable minimum-phase open-loop cases, require $0 < K_P < K_{crit}$.

Fast checks: at $K_P = K_{crit}$, the Nyquist curve passes through -1 and the Bode phase margin is zero.

Common traps: applying this stable-plant Bode rule to an open-loop unstable plant; use Nyquist instead.

7 Nyquist plot and stability

7.1 Ctrl+F keywords

7.1.1 Nyquist, encirclement, critical point, unstable open loop, indkredsning

critical point $-1 + j0$, P , N , Z , unit circle, phase margin

Use this section when open-loop instability or a Nyquist plot appears in the problem.

7.2 Nyquist core formulas

7.2.1 Nyquist curve

$$\Gamma = \{L(j\omega) : \omega \in (-\infty, \infty)\}$$

The plot maps the open-loop frequency response into the complex plane.

7.2.2 Critical point

$$-1 + j0$$

Closed-loop stability depends on encirclements of the critical point.

7.2.3 Encirclement equation

$$Z = P + N$$

P is open-loop right-half-plane poles, N is net clockwise encirclements of -1 , and Z is closed-loop right-half-plane poles.

7.2.4 Closed-loop stability condition

$$Z = 0$$

Choose the encirclement count so no closed-loop poles remain in the right half-plane.

7.3 Margins from Nyquist

7.3.1 Phase margin from unit-circle intersection

$$|L(j\omega_c)| = 1, \quad \gamma_M = 180^\circ + \arg L(j\omega_c)$$

On the Nyquist plot, ω_c is where the curve intersects the unit circle.

7.3.2 Gain margin from negative real-axis crossing

$$L(j\omega_\pi) = -a, \quad A_M = \frac{1}{a}$$

Read $a > 0$ from the negative real-axis intersection.

7.3.3 P gain scaling

$$L_K(j\omega) = K_P L(j\omega)$$

Changing K_P scales the curve radially about the origin.

7.4 Exam recipe: Nyquist stability with unstable open loop

Use when: the plant or loop has right-half-plane poles and Bode margin rules are not enough.

Steps:

1. Count open-loop right-half-plane poles P .
2. Determine the net clockwise encirclements N of $-1 + j0$.
3. Compute $Z = P + N$.
4. Closed loop is stable only if $Z = 0$.
5. For gain changes, scale the Nyquist curve by K_P and repeat the encirclement check.

Fast checks: if $P = 0$, the stable closed loop normally requires no encirclement of -1 .

Common traps: using the Bode positive phase margin rule for an open-loop unstable plant.

7.5 Exam recipe: phase margin from Nyquist plot

Use when: the problem gives a Nyquist curve and asks for phase margin.

Steps:

1. Find where the curve crosses the unit circle.
2. Read the angle $\arg L(j\omega_c)$ from the positive real axis.
3. Compute $\gamma_M = 180^\circ + \arg L(j\omega_c)$.

Fast checks: a point close to $-1 + j0$ has small phase margin.

8 PI-Lead controller design

8.1 Ctrl+F keywords

8.1.1 PI-Lead, PI Lead, lead compensator, phase boost, alpha, tau d, Ni

PI zero, lead zero, lead pole, desired crossover, phase margin, $K_P, \alpha, \tau_i, \tau_d, N_i$

Use this section when the task specifies desired ω_c and γ_M for a PI-Lead controller.

8.2 PI part

8.2.1 PI transfer function

$$C_{PI}(s) = K_P \frac{\tau_i s + 1}{\tau_i s}$$

PI adds an integrator for zero step error and a zero at $-1/\tau_i$.

8.2.2 PI phase at crossover

$$\phi_{PI}(\omega_c) = \arctan(\omega_c \tau_i) - 90^\circ$$

With $N_i = \omega_c \tau_i$, the PI phase contribution is $\arctan(N_i) - 90^\circ$.

8.3 Lead part

8.3.1 Lead transfer function

$$C_{\text{lead}}(s) = \frac{\tau_d s + 1}{\alpha \tau_d s + 1}, \quad 0 < \alpha < 1$$

The lead zero is lower in frequency than the lead pole and adds positive phase.

8.3.2 Lead maximum phase and alpha

$$\phi_{\max} = \sin^{-1} \left(\frac{1 - \alpha}{1 + \alpha} \right), \quad \alpha = \frac{1 - \sin \phi_{\max}}{1 + \sin \phi_{\max}}$$

Smaller α gives more phase lead and more high-frequency gain.

8.3.3 Lead centered at desired crossover

$$\omega_{\max} = \frac{1}{\tau_d \sqrt{\alpha}}, \quad \tau_d = \frac{1}{\omega_c \sqrt{\alpha}}$$

Center the lead phase boost at the desired crossover frequency.

8.4 PI-Lead design equations

8.4.1 PI-Lead controller

$$C(s) = K_P \frac{\tau_i s + 1}{\tau_i s} \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

This is the recurring controller form in the PI-Lead exam task.

8.4.2 Phase-margin design equation

$$\angle G(j\omega_c) + \phi_{PI}(\omega_c) + \phi_{\text{lead}}(\omega_c) = -180^\circ + \gamma_M$$

Use this to compute the required lead phase after choosing the PI zero location.

8.4.3 Crossover magnitude condition

$$|K_P C_{PI,0}(j\omega_c) C_{\text{lead}}(j\omega_c) G(j\omega_c)| = 1$$

$$K_P = \frac{1}{|C_{PI,0}(j\omega_c)| |C_{\text{lead}}(j\omega_c)| |G(j\omega_c)|}$$

$C_{PI,0}$ excludes K_P .

8.5 Exam recipe: PI-Lead design from ω_c and phase margin

Use when: the problem asks for α , τ_d , τ_i , or K_P in a PI-Lead design.

Steps:

1. Choose or read $N_i = \omega_c \tau_i$, then set $\tau_i = N_i / \omega_c$.
2. Compute $\phi_{PI} = \arctan(N_i) - 90^\circ$.
3. Compute required lead phase from the phase-margin equation.
4. Convert required lead phase to α .
5. Set $\tau_d = 1/(\omega_c \sqrt{\alpha})$.
6. Use the magnitude condition to solve K_P .

Fast checks: α must satisfy $0 < \alpha < 1$; τ_i, τ_d, K_P must be positive.

Common traps: forgetting the negative phase from the PI part; using degrees in a calculator set to radians.

9 Design specifications and tuning

9.1 Ctrl+F keywords

9.1.1 Steady-state error, stationær fejl, system type, tuning, specification

step error, ramp error, $K_p, K_v, K_a, \omega_c, \gamma_M, M_p, t_s, K_u, T_u$

Use this section when the task gives a performance specification instead of a controller formula.

9.2 Steady-state error

9.2.1 Final value error formula

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s)$$

The closed loop must be stable before this formula is valid.

9.2.2 Unity-feedback step error

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}$$

For a unit step and unity feedback, low-frequency loop gain directly sets tracking error.

9.2.3 System type and error constants

$$L(s) = \frac{K \prod_i (s - z_i)}{s^q \prod_k (s - p_k)}$$

$$K_p = \lim_{s \rightarrow 0} L(s), \quad K_v = \lim_{s \rightarrow 0} sL(s), \quad K_a = \lim_{s \rightarrow 0} s^2 L(s)$$

The system type q is the number of loop integrators.

9.2.4 Reference error table

Type	step	ramp	parabolic
0	$1/(1 + K_p)$	∞	∞
1	0	$1/K_v$	∞
2	0	0	$1/K_a$

Integral action raises system type and improves polynomial reference tracking.

9.3 Frequency specifications

9.3.1 Crossover speed rule

$$\omega_c \uparrow \implies t_r \downarrow$$

Higher crossover usually gives faster tracking, until limited by phase margin, noise, saturation, and unmodelled dynamics.

9.3.2 Phase margin damping rule

$$\gamma_M \uparrow \implies M_p \downarrow$$

Larger phase margin normally means more damping and lower overshoot.

9.4 P-controller tuning from stability boundary

9.4.1 Critical proportional gain

$$|K_P G(j\omega_\pi)| = 1, \quad \angle G(j\omega_\pi) = -180^\circ$$

At the stability boundary, the loop crosses the critical point.

9.4.2 Magnitude conversion for tuning

$$M_{\text{dB}} = 20 \log_{10} M, \quad M = 10^{M_{\text{dB}}/20}$$

Use this conversion when reading Bode plots and choosing controller gains.

9.5 Exam recipe: steady-state error from diagram or Bode DC gain

Use when: the problem asks for final tracking error, static error, or DC sensitivity.

Steps:

1. Derive $E(s)/R(s)$ or use $S(s) = 1/(1 + L(s))$ for unity feedback.
2. Insert $R(s) = 1/s$ for a unit step.
3. Check closed-loop stability.
4. Compute $e_{\text{ss}} = \lim_{s \rightarrow 0} sE(s)$.
5. If only a Bode plot is given, convert low-frequency dB gain to linear gain first.

Fast checks: type 0 gives finite nonzero step error; type 1 gives zero step error if stable.

Common traps: using the reference-error table for non-unity feedback without deriving the actual E/R .

9.6 Exam recipe: select answer from design specifications

Use when: answer options differ by speed, overshoot, or stability margins.

Steps:

1. Translate time-domain wording to $M_p, t_s, t_r, e_{\text{ss}}$.
2. Translate frequency-domain wording to ω_c, γ_M, A_M .
3. Use ω_c for speed and γ_M for damping.
4. Reject options with unstable poles, negative time constants, or impossible margins.

Fast checks: high K_P normally increases speed but reduces phase margin.

10 Unstable systems

10.1 Ctrl+F keywords

10.1.1 Unstable system, open-loop unstable, right-half-plane pole, unstable

RHP pole, stabilization, inner loop, outer loop, nested loop, Nyquist

Use this section when the plant has a pole with positive real part or the task warns that the open loop is unstable.

10.2 Unstable plant checks

10.2.1 Open-loop unstable pole

$$\exists p_k : \operatorname{Re}(p_k) > 0 \implies G(s) \text{ is open-loop unstable}$$

Stable-plant Bode margin shortcuts do not directly decide closed-loop stability.

10.2.2 Closed-loop stabilization target

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}, \quad 1 + C(s)G(s) = 0$$

The controller must make all roots of the closed-loop characteristic equation lie in the open left half-plane.

10.2.3 Nyquist requirement

$$Z = P + N = 0$$

For an open-loop unstable plant, encirclements must compensate the P right-half-plane open-loop poles.

10.3 Nested-loop design

10.3.1 Inner-loop equivalent plant

$$G_{\text{inner,cl}}(s) = \frac{C_{\text{inner}}(s)G_{\text{inner}}(s)}{1 + C_{\text{inner}}(s)G_{\text{inner}}(s)}$$

Design and verify the inner loop before treating it as part of the outer-loop plant.

10.3.2 Outer-loop transfer

$$L_{\text{outer}}(s) = C_{\text{outer}}(s)G_{\text{outer}}(s)$$

The outer loop uses the stabilized inner closed-loop behavior as its plant model.

10.4 Exam recipe: decide stability for an unstable open-loop plant

Use when: the task combines a right-half-plane pole with K_P , Bode, or Nyquist information.

Steps:

1. Count open-loop right-half-plane poles P .
2. Use Nyquist encirclements or the characteristic polynomial, not the stable-plant Bode shortcut.
3. If using K_P , scale the Nyquist curve or insert K_P in $1 + K_P G(s) = 0$.
4. Require $Z = 0$ or all closed-loop poles in the left half-plane.

Fast checks: a positive phase margin read from a stable-plant formula is not sufficient when $P > 0$.

Common traps: forgetting an open-loop pole at the origin or in the right half-plane.

11 Limited systems and Lag control

11.1 Ctrl+F keywords

11.1.1 Saturation, rate limit, actuator limit, Lag, P-Lead-Lag, beta, begrensning

amplitude limit, rate limit, windup, lag compensator, β, N_i, τ_i

Use this section when the task includes actuator limits or a P-Lead-Lag controller.

11.2 Actuator limitations

11.2.1 Amplitude saturation

$$u_{\text{sat}}(t) = \begin{cases} u_{\text{max}}, & u(t) > u_{\text{max}}, \\ u(t), & u_{\text{min}} \leq u(t) \leq u_{\text{max}}, \\ u_{\text{min}}, & u(t) < u_{\text{min}} \end{cases}$$

Saturation is nonlinear and can invalidate a linear design.

11.2.2 Rate limitation

$$|\dot{u}(t)| \leq R_{\text{max}}$$

Rate limits prevent fast command changes and can reduce effective phase margin.

11.2.3 Integrator windup recognition

$$u_I(t) = \frac{K_P}{\tau_i} \int_0^t e(\xi) d\xi$$

If the actuator saturates while the integral state keeps growing, expect overshoot and slow recovery.

11.3 Lag controller

11.3.1 Lag transfer function

$$C_{\text{lag}}(s) = \frac{\tau_i s + 1}{\frac{\tau_i}{\beta} s + 1}, \quad \beta > 1$$

Lag increases low-frequency gain relative to high-frequency gain without adding an integrator.

11.3.2 P-Lead-Lag controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\frac{\tau_i}{\beta} s + 1}$$

This controller combines phase lead with low-frequency gain shaping.

11.3.3 Lag phase contribution at crossover

$$\phi_{\text{lag}}(\omega_c) = \arctan \left(\frac{N_i(1 - \beta)}{1 + \beta N_i^2} \right), \quad N_i = \omega_c \tau_i$$

This formula is directly tied to the recurring P-Lead-Lag design task.

11.3.4 Lag steady-state error

$$e_{\text{ss}} = \frac{1}{1 + L(0)}$$

Lag gives finite low-frequency gain, so zero step error is not automatic.

11.4 Exam recipe: solve for Lag parameter β

Use when: the problem gives a P-Lead-Lag controller and asks for β or lag phase.

Steps:

1. Read or compute $N_i = \omega_c \tau_i$.
2. Determine the allowed or required lag phase ϕ_{lag} .
3. Insert N_i and ϕ_{lag} in the lag phase equation.
4. Solve for $\beta > 1$.
5. Check the resulting low-frequency gain and phase-margin loss.

Fast checks: lag phase is negative; if the result gives $\beta \leq 1$, it is not a lag compensator.

Common traps: treating Lag like PI; Lag has no pole at the origin and does not guarantee zero step error.

11.5 Exam recipe: recognize limit-induced design problems

Use when: the task asks why a linear design fails on a real actuator.

Steps:

1. Compare requested $u(t)$ with amplitude limits.
2. Compare requested $\dot{u}(t)$ with rate limits.
3. Check whether integral action continues during saturation.
4. Reduce bandwidth, filter reference, or add anti-windup if limits dominate.

Fast checks: large K_P , derivative action, and fast prefilters tend to increase actuator demand.

12 Disturbances, sensitivity, and prefilters

12.1 Ctrl+F keywords

12.1.1 Disturbance, forstyrrelse, sensitivity, prefilter, measurement noise, maaling

output disturbance, input disturbance, $S, T, G_{yd}, G_{ur}, F(s)$, noise

Use this section when the problem asks how disturbances, noise, or prefilters affect the closed loop.

12.2 Sensitivity functions

12.2.1 Sensitivity and complementary sensitivity

$$S(s) = \frac{1}{1 + L(s)}, \quad T(s) = \frac{L(s)}{1 + L(s)}$$

S maps output disturbances and tracking error. T maps reference tracking and measurement-noise transmission.

12.2.2 Input sensitivity

$$G_{ur}(s) = \frac{C(s)}{1 + C(s)G(s)}$$

Use this to estimate control effort and actuator stress.

12.3 Disturbance paths

12.3.1 Output disturbance after plant

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) + \frac{1}{1 + C(s)G(s)}D(s)$$

Additive output disturbances are attenuated by $S(s)$.

12.3.2 Input disturbance before plant

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) + \frac{G(s)}{1 + C(s)G(s)}D(s)$$

The plant shapes the disturbance before feedback attenuation.

12.3.3 Measurement noise

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) - \frac{G(s)C(s)}{1 + C(s)G(s)}N(s)$$

Measurement noise is transmitted through the complementary sensitivity path.

12.4 Prefilters

12.4.1 Reference prefilter

$$Y(s) = \frac{F(s)L(s)}{1 + L(s)}R(s)$$

A prefilter shapes reference response without changing $L(s)$, stability margins, or disturbance rejection.

12.4.2 First-order prefilter

$$F(s) = \frac{1}{\tau_f s + 1}$$

Low-pass prefilters reduce fast reference commands and actuator peaks.

12.5 Exam recipe: derive disturbance transfer function

Use when: the block diagram includes a disturbance input d or noise n .

Steps:

1. Set all other inputs to zero by superposition.
2. Write the summing-junction equations.
3. Solve for $Y(s)/D(s)$, $E(s)/D(s)$, or the requested signal.
4. Express the result using $S = 1/(1 + L)$ and $T = L/(1 + L)$ when possible.
5. Use $s \rightarrow 0$ for constant disturbance checks.

Fast checks: output disturbances usually contain S ; measurement noise usually contains T .

Common traps: treating input and output disturbances as the same transfer path.

12.6 Exam recipe: steady-state disturbance rejection

Use when: the task asks for final output or error due to a constant disturbance.

Steps:

1. Find the disturbance-to-output or disturbance-to-error transfer function.
2. Insert $D(s) = d_0/s$.
3. Check closed-loop stability.
4. Compute the final value with $\lim_{s \rightarrow 0} s(\cdot)$.

Fast checks: high low-frequency loop gain makes $S(0)$ small and improves output-disturbance rejection.

12.7 Exam recipe: prefilter effect

Use when: a task asks whether a prefilter changes margins or disturbance rejection.

Steps:

1. Check whether $F(s)$ is outside the feedback loop.
2. If outside, it changes $R \rightarrow Y$ but not $L(s)$.
3. Stability margins, S , and disturbance rejection are unchanged.
4. Reference speed and overshoot can change.

Common traps: redesigning K_P because of a prefilter; the loop transfer is unchanged.

13 Feed-forward control

13.1 Ctrl+F keywords

13.1.1 Feed-forward, feedforward, disturbance cancellation, forstyrrelseskompensering

measured disturbance, dynamic feed-forward, static feed-forward, properness, $F_d(s)$

Use this section when the disturbance is measured and the problem asks for cancellation before feedback reacts.

13.2 Feed-forward principle

13.2.1 Feedback plus feed-forward input

$$u(t) = u_{fb}(t) + u_{ff}(t)$$

Feed-forward acts directly from reference or measured disturbance and does not need an error first.

13.2.2 Disturbance path model

$$Y(s) = G(s)U(s) + D(s)d(s)$$

G is the manipulated-input path and D is the disturbance path.

13.3 Dynamic disturbance feed-forward

13.3.1 Ideal disturbance cancellation with diagram sign

Disturbance contribution	Ideal feed-forward
$D(s)d(s) + G(s)F_d(s)d(s)$	$F_d(s) = -D(s)/G(s)$
$-D(s)d(s) + G(s)F_d(s)d(s)$	$F_d(s) = D(s)/G(s)$

Choose the sign from the summing junction in the block diagram. The nominal disturbance contribution must become zero.

13.3.2 Disturbance sensitivity with feed-forward

$$G_{yd}(s) = \frac{\sigma_D D(s) + G(s)F_d(s)}{1 + C(s)G(s)}, \quad \sigma_D \in \{-1, +1\}$$

The sign σ_D is read from the disturbance summing junction. Feedback divides the remaining model mismatch by $1 + CG$.

13.3.3 Static feed-forward

$$F_d(0) = -\frac{\sigma_D D(0)}{G(0)}$$

Static feed-forward cancels constant disturbances when the relevant DC gains exist.

13.3.4 Properness condition

$$\deg D(s) - \deg G(s) \leq 0$$

The ideal $F_d = -\sigma_D D/G$ must be proper and stable; otherwise add a realizable filter.

13.3.5 Filtered feed-forward

$$F_{d,\text{filt}}(s) = -\frac{\sigma_D D(s)}{G(s)} Q(s)$$

$Q(s)$ is typically a low-pass filter used to make the feed-forward realizable or less noise-sensitive.

13.4 Exam recipe: design dynamic disturbance feed-forward

Use when: the task gives $G(s)$, $D(s)$, and asks for a feed-forward block.

Steps:

1. Read the sign of the disturbance path from the summing junction.
2. Write the output contribution $Y_d = GU_{\text{ff}} + \sigma_D Dd$.
3. Set $U_{\text{ff}} = F_d d$.
4. Solve $GF_d + \sigma_D D = 0$ to get $F_d = -\sigma_D D/G$.
5. Check that F_d is stable and proper.
6. If not proper or too aggressive, multiply by a low-pass $Q(s)$.

Fast checks: the sign must oppose the disturbance path; $\sigma_D D + GF_d = 0$ for the ideal model.

Common traps: cancelling with the wrong plant path; ignoring improper differentiation in F_d .

13.5 Exam recipe: feed-forward versus feedback

Use when: answer options compare disturbance rejection methods.

Steps:

1. Feed-forward needs a measured disturbance or known reference.
2. Feedback works on unmeasured disturbances but only after error appears.
3. Feed-forward depends strongly on model accuracy.
4. Feedback still determines closed-loop stability margins.

Fast checks: adding feed-forward outside the feedback loop does not by itself change $1 + C(s)G(s)$.